

# How the Two Models Connect

The channels of the OG-Core—CLEWS soft link:  
economic interpretation, model representation, and validation status

Marcelo LaFleur · August 2026 · numbers from the Philippine reference run

# Neither model alone can say who bears an energy transition

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## CLEWS / OSeMOSYS

a least-cost optimiser: chooses what to build and run so energy demand is met at the lowest discounted system cost

- **rich:** technology, operation, emissions, land & water
- **blind:** demand is *fixed* — no price response, no households, no government budget, no distribution

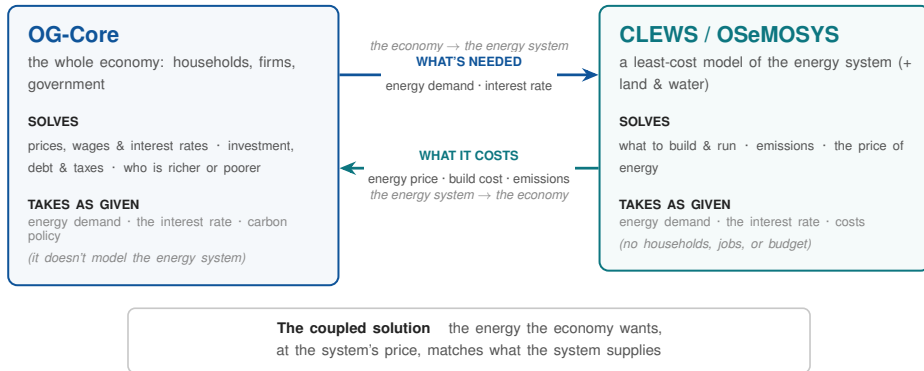
## OG-Core

overlapping-generations general equilibrium: households by age  $S \times$  lifetime income  $J$ ,  $M$  industries, a government that closes its budget each period

- **rich:** behaviour, saving, fiscal policy, incidence, demography
- **blind:** no energy technology — energy is not a priced input to production

Each is detailed exactly where the other is blank, so what crosses the seam differs by direction: **quantities** flow from the economy; **prices and physical outcomes** flow from the energy system.

# Quantities flow one way; prices and outcomes flow the other



*Each model is hollow exactly where the other is deep.*

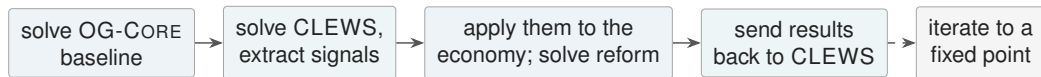
From the economy: energy demand and the interest rate. From the energy system: the energy price, build cost, and emissions. The models stay separate and pass a small set of auditable signals — that asymmetry is the design, not an accident.

# Why a staged soft link, not one model

## The classic hard-link route is foreclosed structurally:

- OG-CORE solves an overlapping-generations *transition path* period by period; OSEMOSYS is one perfect-foresight optimisation over all years at once
- hard links fold the economy in through a scalar welfare objective — and an OLG economy, an equilibrium among cohorts, has none to offer

## So the link runs in stages:



Validated today: the single pass (solid) and the re-solve seam the loop requires. The outer iteration (dashed) is designed, not yet wired.

## The deployed price is a levelized-cost proxy, not a marginal price

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The theoretically clean signal is the marginal cost of electricity — what one more unit would cost the least-cost system to supply; under the competitive reading, the price that quantity would fetch. In this application it is *degenerate*: it binds in scattered years and is near zero elsewhere.

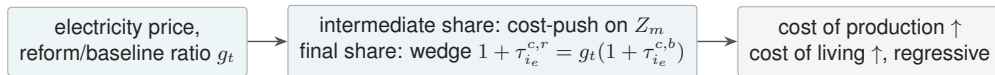
### Four hygiene tests, applied before any marginal cost crosses the seam:

1. above the solver's own reporting resolution;
2. invariant across alternate optima — a price that moves while nothing physical changes is degeneracy, not information;
3. decomposed against any cost parameter that floors it;
4. traceable to a sourced input.

The electricity signal fails the first two, so the deployed price is the **levelized cost of electricity** — dense and robust; the Philippine reform raises it about 5% on average, reaching 16% by 2030.

A fixed point formed with this price is consistency under the selected proxy, not a competitive-equilibrium price.

# Electricity enters through firms' costs and household budgets



- the price splits by *use*: roughly three quarters of Philippine electricity is an industry input (an input–output-calibrated cost-push on industry productivities), one quarter household consumption (a consumption-price wedge); self-use netted so nothing enters twice
- neither side alone is the channel: the wedge is close to output-neutral, the cost-push contractionary — only the composite carries the price into both margins
- with a subsistence floor, poorer households spend a larger energy share  $\Rightarrow$  incidence is regressive even under full revenue recycling

The structural destination — energy as a priced production input — is specified but out of scope.

# The eight channels

| Channel                     | Dir.   | Quantity entering the interface  | Model representation                                       |
|-----------------------------|--------|----------------------------------|------------------------------------------------------------|
| <b>Energy price</b>         | C→O    | reform/baseline cost ratio       | industry cost-push + household wedge                       |
| <b>Public investment</b>    | C→O    | grid & transmission build cost   | $\alpha_I \rightarrow K_g$                                 |
| <b>Capital intensity</b>    | C→O    | fleet capital-cost share         | energy industry's capital share $\gamma_m$                 |
| <b>Health</b>               | C→O    | PM <sub>2.5</sub> emissions path | survival rates + effective labour                          |
| <b>Carbon price</b>         | policy | exogenous price path             | emissions penalty (energy side) + optional household wedge |
| <b>Investment incentive</b> | policy | tax-credit rate                  | user cost of capital                                       |
| <b>Discount rate</b>        | O→C    | equilibrium return               | planning-rate input                                        |
| <b>Energy demand</b>        | O→C    | activity ( $Y_m$ or $C_i$ )      | annual demand path                                         |

The point: nothing crosses the boundary informally. Every exchanged quantity is one of these eight named, guarded channels — four in, two policy, two back.

# Carbon costs enter once

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## Design

- energy side: an emissions penalty changes technology costs and the optimal build
- economy side: a household wedge can price direct carbon content, with revenue recycled
- one price, once — never also read back from the energy model's own implied carbon cost

Provenance checked in the case's own compiled solver input: `EmissionsPenalty` is empty in both the baseline and reform runs.

## In the application (verified)

The Philippine reform solve carries *no* emissions penalty — it is a policy-composition scenario — so **no carbon price acts in the reported coupled numbers**. The flagship run *emits* a \$50/tCO<sub>2</sub> input for the energy model's next solve; the  $-0.021\%$  standalone result is the household-wedge experiment run by itself.



## Three capital mechanisms act on different equations

| Mechanism            | Economic object           | Model entry                 | Interpretation                                                             |
|----------------------|---------------------------|-----------------------------|----------------------------------------------------------------------------|
| Public investment    | grid and transmission     | $\alpha_I \rightarrow K_g$  | productive public capital; debt financing can crowd out private investment |
| Capital intensity    | production factor mix     | energy-industry $\gamma_m$  | changes factor shares, unit cost, and the electricity price                |
| Investment incentive | private capital incentive | tax credit in the user cost | shifts private capital demand toward generation                            |

The capital-share lever and the investment incentive are *not* interchangeable views of one

build-out: they act on different equations, can move energy capital in opposite directions, and the accounting discipline forbids stacking them. Public and private capital are distinct objects — coupling both is complementary, not double counting.

## Raising the capital share sheds capital; the tax credit draws it in

The first-order condition: capital is held in proportion to *revenue*,  $K_E = \gamma_E p_E Y_E / \rho_E$  (share  $\times$  revenue  $\div$  cost of capital). What the lever actually does:

|   | What happens                                                                  | Measured           |
|---|-------------------------------------------------------------------------------|--------------------|
| 1 | the lever raises the capital exponent                                         | $\gamma_E +12.2\%$ |
| 2 | production gets cheaper; zero profit passes the cut into the price            | $p_E -42.8\%$      |
| 3 | cheaper electricity raises demand and output                                  | $Y_E +20.9\%$      |
| 4 | revenue $p_E Y_E$ <i>falls</i> — the price collapse outweighs the output gain |                    |
| 5 | capital is a share of revenue, so capital <i>falls</i>                        | $K_E -22.4\%$      |

The identity  $\hat{K}_E = \hat{\gamma}_E \hat{p}_E \hat{Y}_E$  reproduces the  $-22.4\%$  to within 0.023 pp: the attribution is exact, not a narrative.

**The conclusion, stated plainly:** a higher capital *share* is not capital attraction. The instrument that attracts capital is the investment tax credit — it lowers the cost of capital  $\rho_E$  in the denominator, where  $\gamma_E$  does not appear. Two levers, two different equations, opposite effects on energy capital: never stack them.

Evidence: recomputed from the archived M=8 solved steady states (2026-08-04); a re-run under the re-blessed baseline is queued.

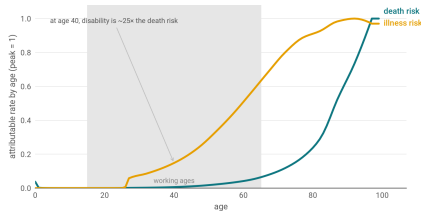
# Cleaner air pays through workers now, and adds retirees later

- CLEWS supplies the  $PM_{2.5}$  path; the GBD supplies the burden and its age profiles (right)
- a calibrated dose-response scales the 2.7% emissions cut to  $\sim 95$  fewer deaths/year (not the 1,165 of one-for-one scaling)
- **mortality** skews elderly  $\rightarrow$  saved lives add retirees  $\rightarrow$  long-run output  $\approx 0$
- **morbidity** is working-age  $\rightarrow$  effective labour rises (+0.19% near-term)

## HEALTH: AGE PROFILES

### Air-pollution death and disability rates, by age

PM<sub>2.5</sub> air-pollution death and illness rates by age - each curve is scaled to its own peak, so this shows the age pattern, not the relative size of death vs illness risk



Source: IHME GBD 2023, ambient particulate-matter burden, Philippines - author's calculations

**So the GDP effect changes sign with the horizon** — gain on the path, cost in the steady state; only an age-structured model produces that split. (The morbidity elasticity is an application assumption.)

# The economy answers back: its return and its demand

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## Discount rate

The equilibrium return  $r_p$  becomes the energy system's planning rate — a harmonisation convention, not a structural equality.

*Still to establish:*

- social rate vs financing cost
- real/nominal and annualisation conventions
- risk and tax adjustments

Both are inert in a single pass by construction; they move the economy only once the loop iterates.

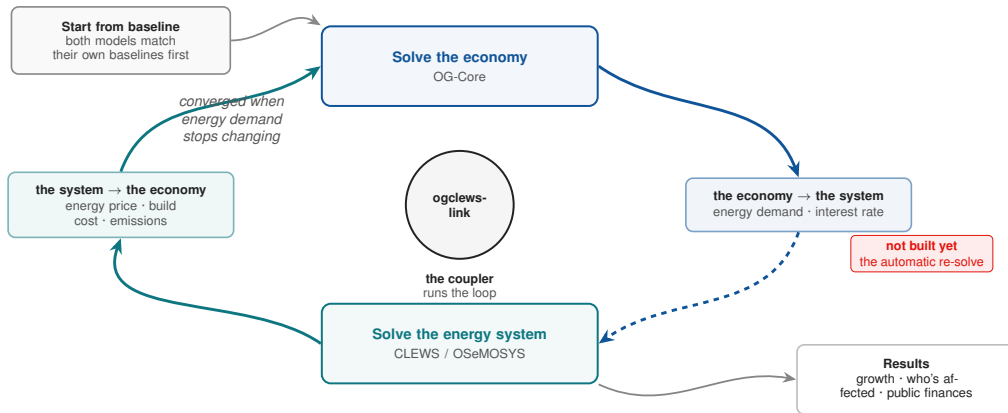
## Energy demand

Sectoral output or consumption scales annual energy-service demand.

*Current evidence:*

- the artifact is produced after the economy solves
- a controlled +10% demand patch re-solves the energy model and raises production +4.3%
- the emitted artifact has not yet passed end to end — that is the outer controller's job

# Closing the loop



Iterate until energy demand stops changing. The re-solve seam is built and tested (a controlled +10% demand patch moved production +4.3%); what remains is the outer iteration and its damping.

## Counting each cost once: five principles

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1. **One carbon price** — an energy-side penalty and an economy-side wedge on the same path; never also inferred from the energy model's implied carbon cost.
2. **Count each energy cost once** — the energy channel partitions electricity by use, self-use netted; a second representation of the same flow is forbidden.
3. **One view of generation capital** — cost-push *or* an investment credit; the capital share is not a third interchangeable view.
4. **Revenue must be routed** — recycle or finance the mechanical wedge revenue, or a hidden fiscal expansion rides along.
5. **Public and private capital are distinct** —  $\alpha_I \rightarrow K_g$  and the firm's own  $K$  touch different objects.

Enforced by guards in the code where a forbidden combination can arise within a channel, and by experiment-composition rules across channels.

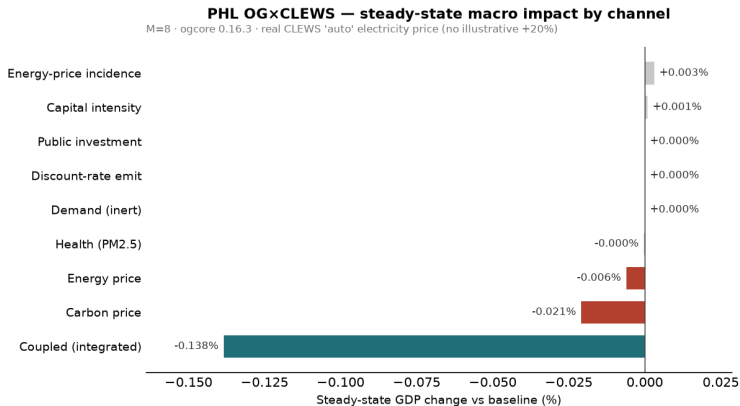
## Three mechanisms drive the Philippine results: price, investment, health



- three mechanisms act: price, investment, health; the emitted signals are inert in one pass (no carbon price acts here)
- the steady-state total is not the sum of the standalone runs — they do not duplicate this treatment
- incidence is regressive under full recycling ( $C_{ss} = -0.092\%$  in the recycled experiment)

| % vs baseline   | steady | $t_0$  | $t_{10}$ |
|-----------------|--------|--------|----------|
| output $Y$      | -0.138 | +0.179 | +0.202   |
| consumption $C$ | -0.087 | +0.066 | +0.385   |

# Each mechanism alone moves output little



Standalone mechanism experiments, steady-state output impact (the figure’s “by channel” title predates the channel/experiment distinction). The energy price is  $\approx 0$  honestly transmitted; carbon ( $-0.021\%$ , the household-wedge experiment) is the largest standalone contraction. Source: the committed golden record.

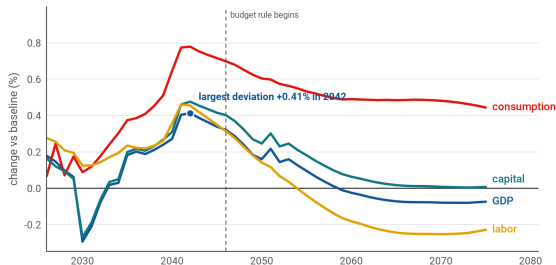


# The coupled economy: gains along the path, costs in the long run

## MACRO TRANSITION

### The economy over time, vs baseline (output, consumption, capital, labor)

% change vs baseline, 2026–2075 · shows how the economy gets there over time



Source: OG-Core x CLEWS coupled model - author's calculations

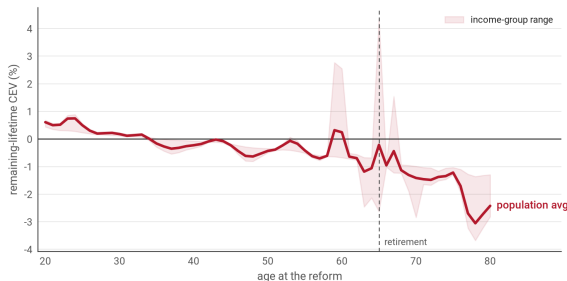
The coupled transition (% vs baseline): output +0.179% near-term, peaking about +0.4% in the early 2040s, then settling toward the -0.138% steady state as the survival gain adds retirees and energy costs persist. The horizon split is the health channel's signature and is invisible to a static model. Figure regenerated from the archived coupled run.

# Who bears it: lifetime welfare by cohort

WELFARE: CEV BY AGE

## Lifetime welfare effect, by current age

Welfare effect by age at the reform, averaged across income groups · ages 20–80



Source: OG-Core x CLEWS coupled model - author's calculations. lifetime CEV (consumption + labor felicity; warm-glow bequest omitted, second-order)

Remaining-lifetime welfare effect by age at the reform, with the income-group range. Younger cohorts gain (they live through the healthier transition); cohorts near and past retirement bear the cost. This cohort resolution is what the overlapping-generations side adds — no representative-agent partner can produce it. Figure regenerated from the archived coupled run.

## Validation is layered, and each layer has a boundary

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| Evidence layer          | What it establishes                                 | Current boundary                               |
|-------------------------|-----------------------------------------------------|------------------------------------------------|
| <b>Transform tests</b>  | shapes, bounds, concordance, fail-fast behaviour    | not an economic calibration                    |
| <b>Controlled runs</b>  | solve completion, direction, identity checks        | some coefficients remain provisional           |
| <b>Golden record</b>    | macro aggregates stable against a committed record  | excludes emitted artifacts and sectoral detail |
| <b>Re-solve seam</b>    | a demand patch survives case copy, solve, read-back | no emitted artifact passed end to end          |
| <b>Closed-loop test</b> | convergence and error propagation                   | outer controller not implemented               |

“Validated” never carries all five meanings at once; each claim in the paper is scoped to its layer.

# Where the coupling stands

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## Established

- the full energy-to-economy pass on real inputs: the scenario's real electricity price, the real Global Burden of Disease data
- controlled validation evidence for the price, capital, and health mechanisms
- the re-solve seam responds correctly to a controlled demand patch
- carbon provenance in the application verified, and the reporting corrected to match

## Next, in order

- the matched cumulative decomposition that reproduces the coupled run
- regression tests for the emitted artifacts
- the capital re-run under the re-blessed baseline
- the damped outer controller — loop closure
- the unit bridge, before any submission

One line: a small, auditable library of economic mechanisms — each signal counted once, each claim scoped to its evidence.